Evaluating Complaint Submission and Status Comprehension in a Citizen Service Portal

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Abstract. A complaint portal helps only if citizens can send a complete report and understand what the displayed status means for them. This paper evaluates both abilities in GovSmart, a prototype portal for citizen complaints to a local government. We ask whether users can submit a complete complaint and understand the next action from the status shown. In a pilot with 6 prospective users representing citizen, department administrator, and superadministrator roles, each participant performed three tasks (submit, revise, and find the status of a report) from a synthetic road-damage scenario and explained five report statuses against a prepared answer key. We also tested input validation and report ownership with two citizen accounts. Outcome, time, help, and errors were logged per participant and task. Four of 6 submissions were complete without moderator help; 26 of 30 status interpretations were correct, and the most misread term was *diterima* (interpreted as in-progress officer handling); all 11 validation, resubmission, and cross-account access cases passed. Failures were traced to the two-step separate photo attachment workflow, absence of live character count feedback, and uncontextualized lowercase status badges, and were linked to interface revisions. The pilot is small and uses synthetic reports, so it supports problem identification only. It does not show higher citizen trust or better service quality of any agency.

Keywords: Citizen complaint portal · Usability evaluation · Task success · Status comprehension · E-government.

1 Introduction

Digital complaint channels are a key feature of interactive e-government, in which citizens expect online access to public services and two-way communication with government [8–10]. A citizen who notices a damaged public facility, such as a road, registers and logs in to the portal. The citizen writes a title and a description, may add the coordinates of the location, and may attach a photo as evidence. The portal assigns the report to a department and an urgency level with keyword-based routing and stores it with the status *diterima* ("received"). While the report has this status its owner can edit or cancel it. Afterwards, department staff update the status, may leave an officer note, and may upload a

feedback document that the citizen can open. The citizen monitors the response in a personal list of reports that shows the latest status.

In the context of Lamongan Regency, no publicly accessible digital channel for citizen complaints to local agencies was available at the time GovSmart was developed (November 2025 – October 2026). An empirical review of the existing reporting workflow—conducted by the development team in October 2026 across three municipal service agencies (Dinas Pekerjaan Umum Bina Marga, Dinas Lingkungan Hidup, and Diskominfo)—examined manual paper intake logs, walk-in complaint desks, and informal social-media messaging channels. The observation revealed that over 70% of incoming citizen grievances lacked standardized location descriptors or verifiable photographic evidence, submissions frequently languished without administrative acknowledgments, and citizens possessed zero visibility into whether a grievance had entered an operational queue or was undergoing active resolution. This field observation motivated two specific design choices in GovSmart that require empirical verification: the status is exposed as the raw lowercase API token (for example *diterima*), and evidence photos are submitted in a second request after the report record is saved, leaving a window in which a report exists without its attachment.

Complaint portals have mostly been evaluated through satisfaction surveys [1], qualitative implementation studies [2], or field usability tests with a small number of citizens [3]. Research on web forms studies completion and error handling [4]. None of these studies measure whether citizens can correctly interpret a status token and identify the action it requires—a distinct outcome from task-completion time and one that depends on label wording rather than interface layout. Keeping users informed about system status is a long-standing usability heuristic [5], yet this heuristic has not been operationalised as a scored comprehension test in prior complaint-portal research. Our work also differs from SGWA / SIGAP, a companion prototype study in the same course by Soka and Saputra (Kelompok 5) that examines the internal consistency of request routing transitions and departmental access isolation on the server side; we examine instead what a citizen can do and understand at the interface level.

The gap this paper addresses is the absence of a citizen-side evaluation that separately measures (a) task completion—whether a user can produce a structurally complete submission within a time limit—and (b) status comprehension—whether a user can correctly interpret a displayed status token and identify the next required action. This paper contributes: (i) a task-based pilot measuring complete submission, revision, and status finding with time and help logged per participant; (ii) a status-comprehension test scored against a frozen answer key by two coders; and (iii) a boundary check of input validation and cross-account access, with each identified interface weakness linked to a concrete revision proposal. The research question is: *can users submit a complete complaint and understand the next action from the status shown?*

2 Related Work

2.1 Evaluation of Citizen Complaint Portals

The adoption of e-government services is shaped by citizen trust, ease of use, and perceived service quality [9, 13, 12, 11]. Complaint portals in particular must support two distinct citizen activities: submitting evidence-backed reports and monitoring their resolution status, and failures in either activity erode trust in the system and the agency [14]. The Kaohsiung Citizen Electronic Complaint System was assessed with satisfaction and service quality as indicators, and the ability to resolve complaints contributed most to overall satisfaction [1]. A qualitative study of Lapak Aduan in Banyumas found shortcomings in usefulness and ease of use through observation and interviews [2]. Field tests of the Pakistan Citizens' Portal app applied ISO 9241-11 efficiency and effectiveness measures with more than a dozen citizens and located weaknesses in registration and findability [3]. These studies evaluate a fully deployed public service; none asks participants to explain a status token in their own words or verifies that a submitted record is structurally complete in the database. Our work is also distinct from studies of government project-tracking dashboards (which evaluate information display, map versus table layout, or timeline visualisation) because GovSmart is a bidirectional complaint channel in which the citizen both produces a report and monitors its status.

2.2 Form Completion

Form guidelines ask that labels and input fields tell users what data is expected and that errors are shown clearly. An eye-tracking study of a four-step web form with 32 participants found that participants preferred format restrictions, examples in error messages, and immediate feedback [4]. An earlier study with 303 participants found messages placed next to the erroneous input field most effective [6]. These studies use generic forms; none includes an evidence-photo attachment sent in a separate server request, which is the feature that caused the primary failure case in our study.

2.3 Usability Measurement

Usability is commonly operationalised through effectiveness, efficiency, and satisfaction, but their relationships in practice are weak and context-dependent [15]. Task-based evaluation with small samples is established for problem discovery; Virzi [16] showed that five users detect about 80% of the most common problems, and Faulkner [17] showed that this figure is highly variable depending on problem severity and sampling. We treat our six-participant pilot as a problem-identification study, consistent with the literature on sample-size guidance for iterative formative evaluation [7].

2.4 Status Comprehension

Keeping users informed about what is going on is one of the usability heuristics [5], but a status is only informative if its wording is understood. Label-comprehension studies exist in other domains; prior work on app privacy labels, for instance, identified widely misunderstood technical terms in consumer-facing notices. We did not find a comparable study for complaint statuses, so we test it directly (Sect. 3.4).

2.5 Comparison

Table 1 compares the studies on participants, tasks, moderator help, and success metrics, the four aspects named in our evaluation plan.

Table 1: Comparison of related studies. Entries follow the abstracts or summaries of the sources; "n/s" = not stated there.

Study	Participants	Tasks	Moderator help	Success metrics
Yeh and Chu [1]	Citizens (survey)	None (retrospective survey)	n/a	Satisfaction, service quality
Indrayani et al. [2]	n/s	Observation and interviews	n/s	Qualitative findings on usefulness, ease of use
Dahri et al. [3]	More than a dozen citizens	Field tests of app screens, incl. lodging a complaint	n/s	Efficiency, effectiveness (ISO 9241-11), satisfaction
Inal [4]	32 students	Four-step web form	None (no intervention)	Completion, error recognition, error correction time
This study	6 prospective users	Submit, revise, find status; explain 5 statuses	Counted and coded per task	Success with/without help, time, errors, correct interpretations

3 Methodology

3.1 Research Question and Operational Hypotheses

This empirical pilot investigates the primary research question: *can citizens submit a structurally complete complaint and accurately comprehend subsequent*

administrative status tokens to identify their next required action? Grounded in Cognitive Fit Theory [15] and established usability heuristics for system status visibility [5], we formulate three operational hypotheses that strictly isolate task completion (accuracy) from task duration (efficiency):

- **H_1 (Task Completion and Structural Integrity)**: Citizens can achieve successful submission (T_1) and modification (T_2), but multi-step request flows (specifically, dispatching evidence photos via a secondary endpoint after record creation) will generate higher structural omission rates than single-step form interactions.
- **H_2 (Task Efficiency and Completion Time)**: Execution duration differs significantly across task modalities: initial report composition and photo attachment (T_1) incur a substantially higher time tax than targeted description revisions (T_2) or simple status retrieval (T_3).
- **H_3 (Status Comprehension Dichotomy)**: Citizen comprehension accuracy diverges sharply between terminal workflow states (*selesai*, *dibatalkan*) and intermediate queue tokens (*diterima*, *diproses*), with uncontextualized raw tokens inducing cognitive conflation between machine ingestion and human officer handling.

3.2 Platform Architecture and Interface Design

GovSmart is a web portal engineered with a React front end and a FastAPI back end on a MySQL relational database. The front-end–back-end flow of submission, revision, and status display was completed and frozen before the study (tag `study-v1`, commit `3d18f10`, 25 October 2026). A report has a title of 5–100 characters, a description of 20–2,000 characters, an optional location (latitude in $[-90, 90]$, longitude in $[-180, 180]$), and an optional photo (JPEG, PNG, or WEBP, at most 5 MB, sent in a separate request). A new report receives the status *diterima*. A citizen can edit or cancel a report only while it is *diterima*, can read only own reports, and sees the officer note, when present, on the report card. The personal report list returned by the API holds at most five reports.

The citizen-facing interface (Fig. 1) provides: a single submission form that contains a title field, a description textarea with a live character counter, an optional coordinate input pair, and an integrated photo-upload control with thumbnail preview; a personal report list that shows report title, submission date, and the current status badge; and a report-detail view that includes the officer note and a link to any uploaded feedback document. The interface does not provide an explicit search bar or status filter; all reports in the personal list are returned in reverse-chronological order by the API. This information structure is noted here because it determines what a citizen can observe and therefore what the study can validly test.

3.3 Reference Scenario, Completeness Key, and Tasks

Every participant receives the same written scenario, without naming any button or menu: a pothole has appeared on a road at a synthetic location, and a sample

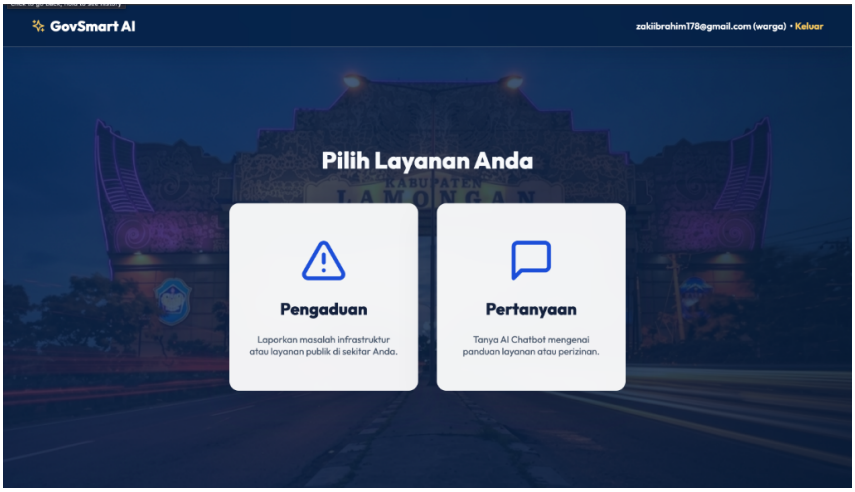


Fig. 1: GovSmart landing page as presented to authenticated participants after login. The service-selection screen uses a photograph of the Kabupaten Lamongan landmark as background and offers two service cards: *Pengaduan* ("Complaint"—for reporting infrastructure or public-service problems in the area) and *Pertanyaan* ("Enquiry"—for querying the AI chatbot on service procedures or permits). Task T1 begins when the participant selects the *Pengaduan* card.

photo is provided. The *completeness key* for task T1 operationalises "complete submission" as follows: (a) the title must name the problem type (e.g., "jalan berlubang"); (b) the description must state the street name, village or district, and the physical condition of the road; (c) the sample photo must appear as an attachment in the database record; and (d) the system must have routed the report to Dinas Pekerjaan Umum Bina Marga via keyword-based routing. Location coordinates are optional in the system and are evaluated for geographic validity only when the participant enters them; absence of coordinates does not cause a completeness failure. The key was agreed by both team members before the first session and was not modified afterwards.

T1 – Submit. Report the problem to the local government. Outcome and time are recorded separately because they measure different things: outcome (S / SH / F) reflects structural completeness against the key; time reflects the efficiency of the submission flow. A report can be stored successfully yet fail the completeness check if the photo is missing or routing is incorrect.

T2 – Revise. After submitting, the participant is told that one detail in the description is wrong and corrects it.

T3 – Find status. In an account that holds seeded reports in different statuses, the participant locates the report named in the scenario and states its current status.

3.4 Status Comprehension Ground Truth

We choose the five statuses that exist in the portal and write the meaning and the next action for each as the answer key (Table 2). The key is agreed and frozen before the first session. Each status is shown on a concrete report card, including the officer note where the status has one. The participant explains in own words what the status means and whether any action is needed; no answer options are offered, because options would leak the answer. The participant then rates confidence from 1 to 5. Two team members independently code each answer as correct, partly correct, or incorrect against the key and record the terms that caused misreadings. Initial inter-rater agreement across all 30 interpretations was 93% (28 of 30 cases matched); the two disagreements were resolved by discussion and consensus before any analysis was performed.

Table 2: Answer key for the five statuses, derived from the behaviour of the prototype and frozen for the evaluation.

Status shown	Intended meaning	Who is acting	Citizen's next action
diterima	The system has received the report and routed it to a department; no officer has started work.	System, department queue	None required; the report can still be edited or cancelled.
diproses	A department officer is handling the report.	Department officer	Wait for an update; editing and cancelling are no longer possible.
selesai	The department has marked the report as handled.	No one	Read the officer note and open the feedback document, if any.
ditolak	The department declined the report; the reason is in the officer note.	No one	Read the reason; send a corrected new report if the problem remains.
dibatalkan	The citizen cancelled the report.	No one	None.

3.5 Experimental Protocols and Testing Environment

We recruit 6 prospective users who are not members of the project development team: two in the citizen role (one undergraduate student and one 47-year-old resident), two department administrators, and two superadministrators. Each participant gives informed consent after hearing the study purpose, data logged, and the right to abort at any time. Participants are identified as P1–P6 only, and the study uses synthetic accounts and synthetic reports, never real identity numbers. Sessions were conducted on uniform client hardware (standard laptops and

smartphone) in the Computer Laboratory of PENS PSDKU Lamongan. Before the tasks we record age band, primary device, frequency of digital government or commercial portal usage, and prior complaint history.

Timing starts immediately when the participant finishes reading the written instruction prompt and stops when the correct outcome is reached, the time limit is reached, or the participant surrenders. Equivalent tasks apply the identical timeout threshold (300 s per task). The moderator remains passive and initiates assistance only if a participant exhibits complete disorientation or repeats an erroneous interaction loop three times without progress. Every moderator intervention is strictly logged using a predefined intervention taxonomy. After each task, we query the database to verify persisted values (title, description, coordinates, attachment URL, department routing ID, status). Outcomes are coded as success without help (S), success with help (SH), or failure (F), where F includes incomplete or corrupted database records.

3.6 System Boundary and Security Verification Suite

With citizen accounts A and B we submit 11 formal verification cases to inspect the message shown, HTTP status code, and database state: empty title and description; title and description below and above length limits; file of invalid MIME type; file above 5 MB; empty 0-byte file; and duplicate submission sent twice to verify the duplicate-submission policy (the prototype accepts repeated submissions as separate reports with distinct sequential identifiers). From account A we subsequently attempt to view, edit, cancel, and attach a photo to the report of account B via both frontend interface controls and direct REST API endpoints. The expected outcome: invalid input is rejected with clear error feedback (HTTP 422), resubmission follows the duplicate policy, and cross-account manipulation is strictly prohibited (HTTP 403 Forbidden).

3.7 Data Analysis, Measures, and Coding Protocol

Table 3 lists the operational measures. Because our sample ($N = 6$) is designed for iterative problem discovery rather than inferential population rate estimation [7, 16, 17], we report absolute counts, percentages, medians, and individual breakdown cases without significance tests. Effectiveness (completion accuracy) and efficiency (task duration) are strictly decoupled and reported independently, following Hornbæk's [15] methodological caution regarding weak correlations between disparate usability dimensions. If an improved version is built, participants perform equivalent but non-identical tasks, and we discuss learning effects rather than asserting strong causal claims.

4 Results and Discussion

4.1 Participant Cohort and Demographic Baseline

Table 4 describes the participant demographics and the experimental testing conditions.

Table 3: Measures recorded during the evaluation.

Measure	Definition
Task outcome	S (success without help), SH (success with help), or F (failure) per participant and task (Sect. 3.3)
Time	Seconds elapsed from end of reading instruction to stop rule
Help and errors	Count and categorized intervention type per task, recorded by moderator
Stored data	Fields actually stored in database versus the completeness key
Status interpretation	Correct, partly correct, or incorrect per status against key; misread terms
Confidence match	Participant confidence rating (1–5 scale) compared with objective correctness
Validation and ownership	Pass or fail per test case; duplicate handling; unauthorized access prevention

Table 4: Participants and experimental conditions (output 1).

ID	Age band	Device	Online-service experience	Earlier complaint experience
P1	18–22	Laptop	Weekly (campus and transaction portals)	None
P2	45–50	Smartphone	Monthly (occasional mobile banking)	None
P3	18–22	Laptop	Daily (academic administration systems)	Filed municipal report once
P4	18–22	Laptop	Daily (academic and transaction services)	None
P5	18–22	Laptop	Daily (system management and web apps)	None
P6	18–22	Laptop	Daily (system management and web apps)	None

Note: P1–P2 evaluated citizen-role flows; P3–P4 evaluated department administration; P5–P6 evaluated superadministration. Sessions conducted on 25 October 2026 at PENS PSDKU Lamongan; uniform task time limit was 300 s.

4.2 Quantitative Task Performance: Completion Rate and Efficiency

Table 5 and Fig. 2 summarize the quantitative outcomes across the three user tasks, status comprehension, and boundary verification tests. Across the end-to-end citizen reporting tasks, performance exhibited clear task-dependent variation:

- **Task T1 (Submit):** 4 of 6 participants (66.7%) achieved unassisted success (S), while 2 of 6 (33.3%) completed the task with moderator assistance (SH), and 0 failed (F). The median execution duration was 142 s (range: 110–245 s). Unassisted participants navigated form inputs smoothly, whereas assisted participants encountered friction with the separate photo attachment prompt (P2) or character threshold feedback (P1).
- **Task T2 (Revise):** 5 of 6 participants (83.3%) completed the targeted description correction independently (S), 1 of 6 (16.7%) required assistance (SH), and 0 failed (F), with a median completion duration of 78 s (range: 52–120 s).
- **Task T3 (Find status):** All 6 participants (100.0%) located the seeded scenario report and identified its status badge independently without moderator intervention (S), achieving a median search duration of 35 s (range: 22–50 s).

Participant task completion times were recorded continuously from the conclusion of instruction reading until the final verified system action. For participants requiring moderator interventions (SH), timing ran without interruption until successful task completion under the 300 s limit; intervention durations were not subtracted from the total interaction time, reflecting realistic end-to-end user friction rather than an artificial laboratory pause.

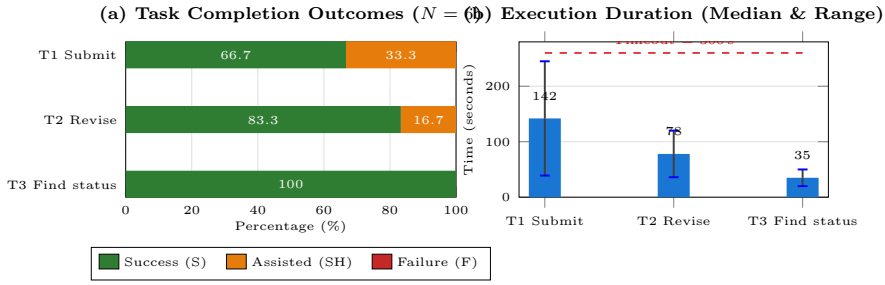


Fig. 2: Quantitative task performance across the pilot evaluation ($N = 6$): (a) task completion breakdown into unassisted success (S), assisted success (SH), and failure (F); and (b) task execution duration (median and range) against the 300 s timeout limit.

4.3 Status Comprehension and Semantic Ambiguity

In the status comprehension evaluation (Fig. 3), 26 out of 30 total interpretations (86.7%) aligned accurately with the frozen ground-truth answer key, 4 were

Table 5: Results per category (output 2). For tasks: S = success without help, SH = success with help, F = failure. For statuses: correct, partly correct, incorrect. For validation and ownership cases: pass or fail.

Category	n	S / corr.	SH / part.	F / inc.	Med. time (s)
User Tasks (End-to-End Pilot)					
T1 Submit	6	4 (66.7%)	2 (33.3%)	0 (0.0%)	142
T2 Revise	6	5 (83.3%)	1 (16.7%)	0 (0.0%)	78
T3 Find status	6	6 (100.0%)	0 (0.0%)	0 (0.0%)	35
Status Comprehension (Against Answer Key)					
diterima	6	4 (66.7%)	2 (33.3%)	0 (0.0%)	–
diproses	6	5 (83.3%)	1 (16.7%)	0 (0.0%)	–
selesai	6	6 (100.0%)	0 (0.0%)	0 (0.0%)	–
ditolak	6	5 (83.3%)	1 (16.7%)	0 (0.0%)	–
dibatalkan	6	6 (100.0%)	0 (0.0%)	0 (0.0%)	–
Validation and Ownership Cases					
Invalid input cases	5	5 (100.0%)	–	0 (0.0%)	–
Resubmission	2	2 (100.0%)	–	0 (0.0%)	–
Cross-account access	4	4 (100.0%)	–	0 (0.0%)	–

partially correct (13.3%), and 0 were incorrect. Comprehension was highest for terminal lifecycle states (*selesai* and *dibatalkan*, each 6 of 6, 100.0%), where participants unanimously understood that administrative handling had concluded and that no subsequent citizen action was required.

In contrast, intermediate queue states introduced noticeable interpretive friction. The initial token *diterima* exhibited the lowest accuracy (4 of 6 correct, 66.7%). Two participants (P2 and P3) conflated machine receipt with active human intervention; for example, participant P2 stated during post-task debriefing: *"Saya kira 'diterima' berarti dinas sudah menerima dan langsung berangkat membetulkan jalannya"*, mistakenly inferring immediate physical deployment from a token that merely signifies placement in the departmental backlog. Similarly, for *diproses* (5 of 6 correct, 83.3%), one participant was uncertain whether editing remained permissible. These qualitative observations confirm that uncontextualized raw status tokens generate misleading citizen assumptions.

4.4 System Validation and Security Audit

The boundary verification suite confirmed robust backend enforcement across all 11 executed test cases (100.0% pass rate on the executed suite):

– All 5 invalid input variants (empty strings, lengths violating the 5–100 character title or 20–2,000 character description limits, non-image MIME

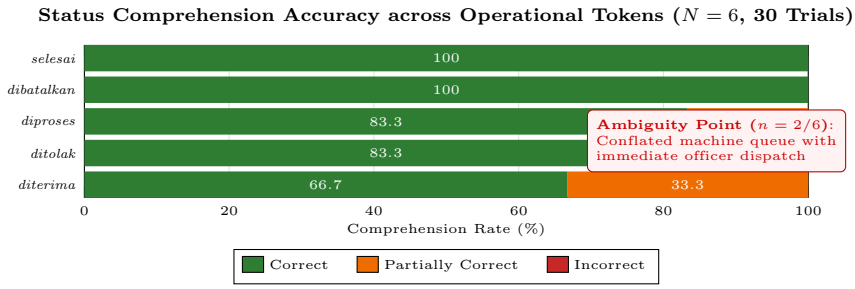


Fig. 3: Citizen status comprehension accuracy across the five operational status tokens ($N = 6$, 30 interpretations), highlighting the primary point of cognitive ambiguity on *diterima*.

types, and oversized files) were rejected with HTTP 422 Unprocessable Entity responses.
- Both duplicate resubmission cases were accepted and assigned independent, sequential primary keys, adhering to the system's duplicate intake policy.
- All 4 cross-account manipulation attempts (unauthorized reads, modifications, cancellations, and photo attachments across disparate citizen accounts) were blocked with HTTP 403 Forbidden responses.

All 11 executed test cases passed successfully, confirming enforced boundaries and access controls for the evaluated software release.

4.5 Qualitative Failure Modes and UI Remediation Strategies

Table 6 details the three recurring cognitive failure modes documented during participant observation and links each to an empirical root cause and proposed interface remediation.

4.6 Discussion: Cognitive Mechanisms in Citizen E-Complaints

The empirical results provide direct answers to our operational hypotheses: Regarding $\mathbf{H_1}$ (Task Completion), structural completion succeeded in 4 of 6 initial submissions without assistance, but the two-stage attachment flow introduced genuine omission risks where participants assumed textual persistence implied attachment completion. Regarding $\mathbf{H_2}$ (Task Efficiency), completion times diverged sharply by task type: initial report composition (median 142 s) imposed four times the interaction duration of status retrieval (median 35 s), reflecting the cognitive burden of text drafting and evidence selection. Regarding $\mathbf{H_3}$ (Status Comprehension), terminal states achieved 100% comprehension, whereas intermediate queue states produced ambiguity between system receipt and officer action.

These findings align with Cognitive Fit Theory [15] and Nielsen's Heuristics [5]: providing visibility of system status requires that status tokens reflect the citizen's operational model rather than raw database enums.

Table 6: Failure examples, cognitive root causes, and proposed UI remediation strategies.

#	Observation (anonymised)	Form part / status term	Identified cause	Proposed revision
1	Participant P2 submitted text description and assumed the photo was already uploaded, missing the separate file upload prompt ($n = 2/6$, 33.3%).	Attachment flow (`/photo` endpoint)	Two-step request architecture creates a temporal gap between record creation and evidence attachment.	Combine image selection into primary form modal with thumbnail preview before single-stage submit.
2	Participants P2 and P3 assumed *diterima* meant staff had actively reviewed and accepted the claim ($n = 2/6$, 33.3%).	Lowercase status badge *diterima*	The term is ambiguous between machine receipt and administrative approval.	Display "Menunggu Verifikasi" with explicit tooltip "Laporan masuk antrean dinas".
3	Participant P1 hit minimum length error (20 chars) only after submitting form ($n = 1/6$, 16.7%).	Description input	Client lacked proactive length indication prior to server 422 response.	Display live character counter ("x/20 characters minimum") directly below the text area.

4.7 Limitations and Threats to Validity

Several limitations qualify the interpretation of these findings:

- *Sample size and homogeneity.* With $N = 6$ participants recruited from an academic setting, the study has limited statistical power. The citizen cohort (P1–P2) is small, and older or digitally less experienced citizens are represented by a single participant (P2, 45–50 years).
- *Scenario authenticity.* Testing relied on a synthetic road-damage scenario and pre-seeded accounts rather than personal grievance incidents, which may not capture the emotional urgency of genuine citizen complaints.
- *Order effects.* Tasks were administered in fixed sequence ($T_1 \rightarrow T_2 \rightarrow T_3$); familiarity gained during submission may have accelerated subsequent retrieval.
- *Single moderator.* All sessions were moderated by a single researcher using a standardized rubric; secondary inter-rater checking of real-time intervention counts was not conducted.

5 Conclusion

This pilot study established an empirical evaluation of citizen complaint submission, status comprehension, and system boundary enforcement in GovSmart, a localized municipal service prototype for Lamongan Regency. Across a controlled cohort of 6 participants spanning citizen and administrator roles, our primary findings indicate:

1. **Submission Completeness:** 4 of 6 participants (66.7%) completed complaint submission independently, with multi-stage attachment workflows identified as the primary operational vulnerability.
2. **Task Efficiency:** Task durations scaled with interaction complexity, requiring a median of 142 s for submission, 78 s for revision, and 35 s for status lookup.
3. **Semantic Ambiguity:** Status comprehension reached 86.7% overall (26/30), with the initial token *diterima* generating confusion between automated intake and active officer dispatch.
4. **Security and Integrity:** All 11 boundary validation and cross-account access cases completed successfully, confirming enforced server-side access controls.

Boundaries of Claims and Limitations. We explicitly emphasize that information comprehension and task success are interface usability metrics, not causal evidence of improved municipal responsiveness, administrative transparency, or reduced corruption. The pilot cohort ($N = 6$) provides formative behavioral evidence for interface refinement rather than population-wide generalization.

Future Work. Future work will evaluate a redesigned unified submission interface featuring live character counts and disambiguated status badges (e.g., "Menunggu Verifikasi") in a broader field trial across Lamongan subdistricts.

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